

QUARTERLY BUSINESS PERFORMANCE REPORT
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Q4 2025 (October - December)

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Prepared For: Board of Directors & Risk Committee
Prepared By: Analytics & Business Intelligence Team
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EXECUTIVE SUMMARY
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The Credit Risk Prediction API and XGBoost-based underwriting system completed its first full quarter of production deployment in Q4 2025. The system processed 45,231 loan applications with total origination volume of \$89.4M, representing a 23% increase in approval efficiency compared to the legacy underwriting system. Most notably, the model prevented an estimated \$1.2M in potential defaults through improved risk selection, while operational cost savings reached \$450,000 from automated decisioning.

KEY METRICS AT A GLANCE:

Metric	Q4 2025	Q3 2025	QoQ Change	Target	Status
Total Applications	45,231	38,942	+16.2%	42,000	▲ Exceeded
Approval Rate	71.1%	68.3%	+2.8pp	70%	▲ Exceeded
Origination Volume	\$89.4M	\$76.2M	+17.3%	\$85M	▲ Exceeded
Avg Loan Size	\$19,755	\$18,982	+4.1%	\$19,500	▲ Exceeded
Weighted Avg PD	6.23%	6.89%	-0.66pp	<7.0%	▼ Improved
30+ Day Delinquency	2.34%	2.89%	-0.55pp	<2.5%	▼ Improved
90+ Day Default Rate	1.12%	1.45%	-0.33pp	<1.3%	▼ Improved
Estimated Loss Rate	0.84%	1.09%	-0.25pp	<1.0%	▼ Improved
API Uptime	99.97%	99.94%	+0.03pp	99.95%	▲ Exceeded
Avg Latency (p95)	78ms	92ms	-14ms	<100ms	▲ Exceeded

The system successfully balanced risk reduction with business growth, demonstrating the effectiveness of the XGBoost model in production. No significant model drift or performance degradation was observed during the quarter.

1.0 MODEL PERFORMANCE DEEP DIVE

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1.1 Predictive Performance Metrics

The XGBoost model (version 2.1.0, trained February 2025) maintained strong predictive performance throughout Q4 2025. Performance was evaluated using three methodologies:

- Real-time performance monitoring (available immediately)
- Delayed label analysis (90-day lag for default confirmation)
- Backtesting against Q4 outcomes (full quarter results available Jan 15)

PRIMARY METRICS (on Q4 production data, n=45,231):

Metric	Value	95% CI	Benchmark	Status
AUC-ROC	0.958	[0.951, 0.965]	0.950	▲ Superior
AUC-PR	0.872	[0.861, 0.883]	0.850	▲ Superior
Accuracy	93.36%	[92.89%, 93.83%]	92.50%	▲ Superior
Precision (Default)	94.80%	[93.21%, 96.39%]	90.00%	▲ Superior
Recall (Default)	74.10%	[71.23%, 76.97%]	75.00%	▼ Below Target
F1 Score (Default)	83.12%	[81.45%, 84.79%]	82.00%	▲ Superior
Specificity	96.20%	[95.81%, 96.59%]	95.00%	▲ Superior
Negative Predictive Value	97.45%	[97.01%, 97.89%]	97.00%	▲ Superior
Matthews Correlation	0.742	[0.721, 0.763]	0.720	▲ Superior

ANALYSIS:

The model significantly exceeds targets on most metrics. The lower-than-target recall (74.10% vs 75.00% target) indicates the model is slightly conservative - it misses 0.9% more defaults than desired. However, the exceptional precision (94.80%) means that when the model predicts default, it's correct 95% of the time. This precision-recall tradeoff is acceptable given the cost of defaults (\$5,000 avg loss) vs cost of false rejects (lost revenue ~\$1,000 per false reject).

1.2 Calibration Analysis

Model calibration (whether predicted probabilities match observed default rates) is critical for risk-based pricing and capital allocation.

CALIBRATION CURVE DATA (Q4 2025):

PD Decile	Predicted PD	Observed Default Rate	Calibration Error	Well-Calibrated?
1 (0-1%)	0.42%	0.38%	-0.04%	✓ Yes

2 (1-2%) 1.51% 1.49% -0.02% ✓ Yes
3 (2-3%) 2.48% 2.52% +0.04% ✓ Yes
4 (3-5%) 3.95% 4.01% +0.06% ✓ Yes
5 (5-7%) 5.98% 6.12% +0.14% ✓ Yes
6 (7-9%) 7.89% 8.04% +0.15% ✓ Yes
7 (9-11%) 9.95% 10.23% +0.28% ⚠ Slight over-confidence
8 (11-13%) 11.98% 12.45% +0.47% ⚠ Over-confident
9 (13-15%) 13.92% 14.78% +0.86% ✗ Over-confident
10 (15%+) 18.45% 19.89% +1.44% ✗ Over-confident

FINDINGS:

The model is well-calibrated for PD < 9% (Deciles 1-7). For higher risk tiers (PD > 9%), the model exhibits systematic under-prediction of default risk (actual defaults exceed predictions by 0.28-1.44%). This has been escalated to the Model Risk team for investigation. Recommended actions:

- Recalibrate high-risk predictions using isotonic regression
- Consider separate model for high-risk segment
- Add 15% risk buffer to PD estimates > 10% until recalibration complete

1.3 Feature Stability Analysis

Population Stability Index (PSI) measures feature distribution shifts between training (Jan-Dec 2024) and production (Q4 2025).

Feature	PSI	Stability Rating	Action Required
-----	-----	-----	-----
loan_int_rate	0.042	Stable	None
person_income	0.038	Stable	None
person_emp_length	0.035	Stable	None
loan_amnt	0.051	Stable	None
person_age	0.029	Very Stable	None
loan_percent_income	0.048	Stable	None
cb_person_cred_hist_length	0.044	Stable	None
loan_grade distribution	0.067	Slightly Unstable	Monitor
person_home_ownership	0.059	Slightly Unstable	Monitor
loan_intent distribution	0.072	Slightly Unstable	Review

OVERALL POPULATION STABILITY INDEX: 0.051 (Stable)

- PSI < 0.10: No action needed
- 0.10 ≤ PSI < 0.20: Investigation recommended
- PSI ≥ 0.20: Model retraining required

CONCLUSION: No significant population drift detected. Continue monitoring loan_intent distribution (increase in MEDICAL and DEBTCONSOLIDATION intents, decrease in VENTURE intents).

1.4 SHAP Analysis - Top Drivers

SHAP (SHapley Additive exPlanations) values identify which features most influenced individual predictions in Q4.

TOP 5 FEATURES BY MEAN |SHAP|:

Rank	Feature	Mean SHAP	Direction	Business Insight
1	loan_int_rate	0.124	Positive	Higher rates → Higher default risk. Each 1% rate increase → +0.8pp PD
2	loan_percent_income	0.108	Positive	DTI > 0.45 is critical threshold. 3x risk above 0.45
3	person_emp_length	0.096	Negative	Employment < 1 year = 2.5x risk. Every +2 years → -0.5pp PD
4	prior_default_flag	0.089	Positive	Prior default within 36 months → 4.2x base risk
5	person_income	0.082	Negative	Income < \$40k = high risk. \$100k+ = low risk

INTERACTION EFFECTS DETECTED:

- loan_int_rate × prior_default_flag: Interaction effect +0.034 SHAP
- person_emp_length × person_age: Interaction effect -0.028 SHAP
- loan_percent_income × loan_amnt: Interaction effect +0.022 SHAP

These interactions explain 7.4% of total model variance and are being considered for explicit feature engineering in the next model version.

2.0 BUSINESS OPERATIONAL METRICS

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2.1 Application Volume & Trends

DAILY APPLICATION VOLUME (Q4 2025):

Month	Total Apps	Avg Daily	Peak Day	Low Day	Std Dev
October	14,892	480	723	312	118
November	13,456	449	698	289	124
December	16,883	545	892	341	156
Total	**45,231**	**491**	**892**	**289**	**141**

SEASONALITY OBSERVATIONS:

- December peak (+21% vs monthly average) driven by holiday spending
- Lowest volume: Thanksgiving week (-32% vs average)
- Weekend volume: 68% of weekday average

HOURLY DISTRIBUTION:

- Peak hours: 10 AM - 2 PM (42% of daily volume)
- Evening hours: 6 PM - 9 PM (23% of volume)
- Overnight (12 AM - 6 AM): 8% of volume

APPLICATION SOURCE CHANNELS:

Channel	Volume	Approval Rate	Avg PD	Avg Loan Size
Direct Web	18,234 (40.3%)	74.2%	5.87%	\$21,450
Mobile App	15,678 (34.7%)	72.8%	6.12%	\$19,230
Partner/Affiliate	7,892 (17.4%)	63.4%	7.45%	\$16,890
Phone/Agent	2,345 (5.2%)	69.1%	6.78%	\$18,340
In-Branch	1,082 (2.4%)	71.2%	6.34%	\$19,120

INSIGHT: Partner channel has lower approval rates and higher risk profiles, suggesting different customer acquisition quality. Recommend adjusting partner incentive structure or implementing channel-specific risk models.

2.2 Approval & Decision Analysis

DECISION BREAKDOWN BY RISK TIER:

Tier	Applications	Approved	Rejected	Manual Review	Approval Rate
A+	2,891 (6.4%)	2,834	12	45	98.0%
A	6,783 (15.0%)	6,444	102	237	95.0%
B+	7,234 (16.0%)	6,511	217	506	90.0%
B	8,143 (18.0%)	6,107	543	1,493	75.0%
C+	7,012 (15.5%)	3,506	1,753	1,753	50.0%
C	5,879 (13.0%)	1,470	2,205	2,204	25.0%
D+	3,617 (8.0%)	362	1,809	1,446	10.0%
D	2,713 (6.0%)	136	1,900	677	5.0%
E	859 (1.9%)	0	859	0	0.0%
Total	**45,231**	**27,370**	**9,400**	**8,461**	**71.1%**

AUTOMATED vs MANUAL DECISIONS:

Decision Type	Count	Percentage	Avg Processing Time
Fully Automated - Approve	21,234	46.9%	1.2 seconds
Fully Automated - Reject	7,834	17.3%	0.9 seconds
Manual Review - Approved	6,136	13.6%	4.2 hours
Manual Review - Rejected	1,566	3.5%	3.8 hours
Manual Review - Pending	759	1.7%	In progress
Referred (incomplete data)	1,702	3.8%	N/A

MANUAL REVIEW PERFORMANCE:

- Average manual review time: 4.1 hours (target: < 6 hours) ✓

- Overturn rate (automated decision changed): 12.3%
- Approve-to-reject overturn: 4.2%
- Reject-to-approve overturn: 8.1%
- Average loss on overturns (reject-to-approve): \$2,340 per loan
- Benefit from overturns (additional revenue): \$4.2M

INSIGHT: Manual reviewers correctly identified 8.1% of applicants that should have been approved despite automated rejection. The additional revenue from these overturns (\$4.2M) significantly exceeds losses from defaults on overturns. Recommend recalibrating model to be less conservative.

2.3 Geographic Performance

TOP 10 STATES BY APPLICATION VOLUME:

Rank	State	Volume	Approval Rate	Avg PD	Avg Loan Size	Default Rate
-----	-----	-----	-----	-----	-----	-----
1	California	6,892	68.2%	6.89%	\$22,450	1.34%
2	Texas	5,234	72.1%	6.12%	\$18,920	1.08%
3	Florida	4,123	70.4%	6.45%	\$19,340	1.21%
4	New York	3,892	67.8%	7.01%	\$21,890	1.42%
5	Illinois	2,567	73.2%	5.89%	\$18,450	0.98%
6	Pennsylvania	2,234	74.1%	5.67%	\$17,890	0.92%
7	Ohio	2,089	75.2%	5.45%	\$16,230	0.87%
8	Georgia	1,987	69.8%	6.78%	\$18,670	1.18%
9	North Carolina	1,834	74.5%	5.78%	\$17,450	0.94%
10	Michigan	1,756	73.8%	5.92%	\$16,890	0.96%

STATE-LEVEL INSIGHTS:

- Highest risk: New York, California (PD > 6.8%)
- Lowest risk: Ohio, Pennsylvania (PD < 5.7%)
- Regulatory considerations: New York and California have additional disclosure requirements. Compliance rate: 99.2%.

3.0 FINANCIAL PERFORMANCE

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3.1 Revenue & Profitability

ORIGINATION VOLUME BY MONTH:

Month	Loan Count	Total Originated	Avg Loan Size	Origination Fees	Interest Income (Est)
-----	-----	-----	-----	-----	-----
October	8,912	\$28.4M	\$19,120	\$568K	\$1.42M

November	8,234	\$27.6M	\$19,890	\$552K	\$1.38M	
December	10,224	\$33.4M	\$20,450	\$668K	\$1.67M	
Total	**27,370**	**\$89.4M**	**\$19,755**	**\$1.788M**	**\$4.47M**	

Note: Interest income represents estimated first-year interest. Total loan lifetime interest estimated at \$22.3M over weighted average loan term of 36 months.

PROFITABILITY ANALYSIS:

Metric	Q4 2025	Q3 2025	Change	
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Gross Interest Income (Lifetime)	\$22.3M	\$19.0M	+\$3.3M	
Origination Fees	\$1.788M	\$1.524M	+\$264K	
Total Gross Revenue	**\$24.088M**	**\$20.524M**	**+\$3.564M**	
Expected Credit Losses (ECL)	\$0.75M	\$0.83M	-\$80K	
Operating Expenses	\$1.45M	\$1.68M	-\$230K	
Marketing Costs	\$1.89M	\$1.62M	+\$270K	
Technology Costs	\$0.42M	\$0.41M	+\$10K	
Compliance/Legal	\$0.31M	\$0.29M	+\$20K	
Total Expenses	**\$4.82M**	**\$4.83M**	**-\$10K**	
Net Profit (Lifetime)	**\$19.268M**	**\$15.694M**	**+\$3.574M**	
Net Profit Margin	80.0%	76.5%	+3.5pp	

ROI ANALYSIS:

- Return on Assets (ROA): 21.6% (annualized)
- Return on Equity (ROE): 34.2% (assuming 2:1 leverage)
- Customer Acquisition Cost (CAC): \$89 per funded loan
- Lifetime Value (LTV): \$880 per customer
- LTV/CAC Ratio: 9.9x (excellent, target > 5x)

3.2 Expected vs Actual Losses

LOSS ANALYSIS BY RISK TIER (annualized):

Tier	Originations	ECL Rate	Actual Loss Rate (Annualized)	Variance	Performance	
----- ----- ----- ----- -----						
A+	\$5.8M	0.15%	0.12%	-0.03%	▲ Better	
A	\$13.5M	0.35%	0.31%	-0.04%	▲ Better	
B+	\$14.2M	0.70%	0.68%	-0.02%	▲ Better	
B	\$17.8M	1.20%	1.18%	-0.02%	▲ Better	
C+	\$13.4M	2.10%	2.23%	+0.13%	▼ Worse	
C	\$8.9M	3.50%	3.78%	+0.28%	▼ Worse	
D+	\$4.2M	5.50%	6.01%	+0.51%	▼ Worse	
D	\$1.6M	8.00%	8.89%	+0.89%	▼ Worse	
E	\$0	15.00%	N/A	N/A	N/A	
Total	**\$79.4M**	**1.18%**	**1.23%**	**+0.05%**	**Slightly Worse**	

OBSERVATION: Actual losses slightly exceed expectations for Tiers C and above (subprime segments). This aligns with the calibration error identified in Section 1.2. Recommend:

- Increase risk reserves for Tier C+ by 15%
- Adjust pricing for Tier C and D by +2-3%
- Enhanced collection efforts for subprime portfolio

TOTAL PROVISION FOR CREDIT LOSSES (PCL):

- Opening balance (Oct 1): \$2.1M
- Q4 provision: \$0.75M
- Write-offs (actual defaults): \$0.62M
- Recoveries: \$0.08M
- Closing balance (Dec 31): \$2.31M

PCL coverage ratio (PCL / total outstanding): 3.9% (within policy target of 3-5%)

3.3 Cost Savings from Automation

DIRECT COST COMPARISON:

Cost Category	Legacy Manual System	Automated API System	Savings	% Reduction
Underwriter salaries	\$385,000	\$95,000	\$290,000	75.3%
Review time cost	\$178,000	\$42,000	\$136,000	76.4%
Error correction	\$45,000	\$12,000	\$33,000	73.3%
Overtime/bonus	\$32,000	\$8,000	\$24,000	75.0%
Training	\$18,000	\$4,000	\$14,000	77.8%
Quality assurance	\$42,000	\$15,000	\$27,000	64.3%
Total	**\$700,000**	**\$176,000**	**\$524,000**	**74.9%**

Note: Automated system costs include model hosting, API infrastructure, and monitoring (estimated at \$0.0039 per API call × 45,231 calls = \$176K).

INDIRECT COST SAVINGS:

- Reduced customer acquisition time (faster approval → higher conversion)
- Lower complaint volume (consistency of automated decisions)
- Decreased regulatory risk (auditable, consistent decisions)
- Eliminated paper processing costs (\$12K saved)

TOTAL ANNUALIZED SAVINGS: ~\$1.8M (when scaled to full year)

4.0 API & TECHNICAL PERFORMANCE

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4.1 API Reliability Metrics (Q4 2025)

Metric	Q4 Value	Target	Status
Uptime (excluding planned maintenance)	99.97%	99.95%	✓ Exceeded
Number of incidents	2	<5	✓ Exceeded
Mean time to detect (MTTD)	4.2 min	<10 min	✓ Exceeded
Mean time to recover (MTTR)	18 min	<30 min	✓ Exceeded
Error rate (5xx responses)	0.03%	<0.10%	✓ Exceeded
Timeouts (< 30s)	0.001%	<0.01%	✓ Exceeded

INCIDENT REPORT:

Incident #1: November 12, 2025 - Database connection pool exhaustion

- Duration: 7 minutes (03:14 - 03:21 UTC)
- Impact: 234 requests failed (0.5% of daily volume)
- Root cause: Inadequate connection pool size during traffic spike
- Resolution: Increased pool size and added connection retry logic
- Prevention: Connection pool monitoring alert added

Incident #2: December 24, 2025 - Third-party credit bureau API latency

- Duration: 11 minutes (14:23 - 14:34 UTC)
- Impact: 412 requests delayed (average +2.3 seconds)
- Root cause: External vendor performance degradation
- Resolution: Circuit breaker pattern prevented system-wide impact
- Prevention: Added fallback model (trained on bureau-less features)

4.2 Performance & Scalability

LATENCY DISTRIBUTION (p50, p95, p99 in milliseconds):

Endpoint	p50	p95	p99	Max
/health	2ms	5ms	8ms	15ms
/docs	12ms	23ms	31ms	48ms
/predict (standard)	32ms	78ms	112ms	189ms
/predict (with explanation)	54ms	112ms	156ms	234ms
/batch_predict	156ms	289ms	423ms	612ms

LOAD TEST RESULTS (performed December 28, 2025):

Concurrent Users	Requests/sec	p95 Latency	Error Rate	CPU Usage
10	45	48ms	0.00%	12%
50	210	67ms	0.00%	28%
100	415	89ms	0.00%	45%
250	980	145ms	0.00%	72%

| 500 | 1,780 | 234ms | 0.02% | 91% |
| 1000 | 2,450 | 456ms | 1.23% | 100% |

SUSTAINABLE THROUGHPUT: ~1,200 requests/second at p95 < 200ms
CURRENT AVERAGE LOAD: ~0.6 requests/second (well below capacity)
SCALING RECOMMENDATION: Current infrastructure adequate for 200x current volume

4.3 Data Quality & Completeness

INPUT DATA QUALITY METRICS:

Feature	Missing Rate	Outlier Rate	Invalid Values	Data Type Errors
person_age	0.02%	0.31%	0.01%	0.00%
person_income	0.15%	0.89%	0.08%	0.01%
person_emp_length	1.23%	0.12%	0.23%	0.00%
loan_amnt	0.00%	0.45%	0.00%	0.00%
loan_int_rate	0.34%	0.23%	0.05%	0.00%
loan_percent_income	0.18%	1.12%	0.12%	0.00%
cb_person_cred_hist_length	0.45%	0.08%	0.04%	0.00%

DATA QUALITY SCORE: 98.7% (target > 98%) ✓

TOP DATA ISSUES (by volume):

1. Employment length missing or "unknown" (1.23%)
2. Credit history length missing (0.45%)
3. Loan interest rate missing (0.34%)
4. Income outliers requiring winsorization (0.89%)

All missing data handled via imputation strategies defined in data_prep.py.

5.0 COMPLIANCE & REGULATORY

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5.1 Fair Lending Analysis

DISPARATE IMPACT TESTING (adverse impact ratio by protected class):

Protected Class	Approval Rate	Adverse Impact Ratio	Disparity Found?	Action Required
Race - White (reference)	72.1%	1.00	N/A	N/A
Race - Black/African American	68.4%	0.949	No (above 0.80 threshold)	Monitor
Race - Hispanic/Latino	69.2%	0.960	No	Monitor
Race - Asian	73.4%	1.018	No	None

Race - Other	70.1%	0.972	No	None
Gender - Male (reference)	71.8%	1.00	N/A	N/A
Gender - Female	70.2%	0.978	No	None
Gender - Non-binary	69.8%	0.972	No	None
Age - 25-35 (reference)	73.2%	1.00	N/A	N/A
Age - Under 25	62.1%	0.848	No (below 0.80 would be violation)	Monitor closely
Age - 36-50	72.8%	0.995	No	None
Age - 51-65	71.2%	0.973	No	None
Age - Over 65	67.4%	0.921	No	None

FINDING: No ECOA violations detected (all adverse impact ratios > 0.80). However, approval rates for applicants under 25 are significantly lower (62.1%). This is driven by objective risk factors (limited credit history, lower income), not prohibited bias. Recommendation: Document business justification and monitor quarterly.

5.2 Adverse Action Notices

Metric	Q4 2025	Regulatory Requirement	Compliance
----- ----- ----- -----			
AAN sent for denials	9,400 (100%)	100%	✓ Compliant
AAN sent within 30 days	9,398 (99.98%)	100%	⚠ 2 Late
AAN with specific reasons	9,400 (100%)	Required	✓ Compliant
AAN with right to dispute	9,400 (100%)	Required	✓ Compliant
Consumer complaints received	23	N/A	Monitor

ACTION ITEM: Investigate 2 late AANs (delays of 32 and 34 days). Root cause identified as mailroom processing delay. Resolution: Implement electronic delivery as default, mail as exception.

5.3 Regulatory Examinations

No regulatory examinations occurred in Q4 2025. Next scheduled examination:

- CFPB: Q2 2026
- State banking departments: Various dates (tracking maintained by Compliance)

INTERNAL AUDIT FINDINGS (December 2025):

Finding	Severity	Status	Remediation Date
----- ----- ----- -----			
Model documentation outdated (version 2.0 vs deployed 2.1)	Medium	Resolved	Jan 5, 2026
Missing approval overrides logging for 12 loans	High	Resolved	Dec 28, 2025
Data retention policy not applied to test environment	Low	In Progress	Jan 30, 2026

All high-severity findings remediated within 5 business days.

6.0 RISK & MITIGATION

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6.1 Model Risk Assessment

MODEL RISK TIER (per SR 11-7): HIGH

Rationale: Model directly influences credit decisions with material financial impact.

KEY MODEL RISKS IDENTIFIED:

Risk Category	Description	Severity	Mitigation	Status
Calibration drift	Over-confidence in high-risk predictions (identified in 1.2)	High	Recalibration by Feb 15, 2026	In Progress
Concept drift	Potential shift in default drivers post-COVID	Medium	Monthly monitoring; quarterly retraining	Active
Data quality	Missing employment history (1.23%)	Low	Imputation; upstream data fixes	Monitoring
Implementation	API error handling gaps	Medium	Enhanced error handling (Dec 2025)	Resolved
Overfitting	Validation vs test gap of 1.2%	Low	Regularization; out-of-time testing	Acceptable

MITIGATION STATUS: 1 resolved, 4 active monitoring, 1 in progress (recalibration)

6.2 Operational Risks

Risk	Probability	Impact	Mitigation	Residual Risk
API outage	Low (0.03% actual)	High	Multi-zone deployment; circuit breakers	Low
Model serving failure	Very Low (0.01%)	High	Fallback rule-based model	Low
Data breach	Very Low	Critical	Encryption; access controls; monitoring	Very Low
Third-party (credit bureau) outage	Medium	Medium	Fallback model (bureau-less)	Medium-Low
Regulatory fine for AI bias	Low	High	Fair lending testing; documentation	Low

No operational risk events occurred in Q4 2025.

6.3 Concentration Risk

PORTFOLIO CONCENTRATION:

Category	Concentration	Policy Limit	Status

Tier A+B (Prime)	47.6%	No limit	Acceptable
Tier C (Near Prime)	33.2%	35% max	Within limit
Tier D (Subprime)	8.1%	10% max	Within limit
Tier E (Deep Subprime)	0%	3% max	Within limit
Top state (California)	18.2%	20% max	Within limit
Top 3 states	38.9%	40% max	Within limit
Partner channel	17.4%	25% max	Within limit

No concentration limit breaches. Recommend reducing California exposure from 18.2% to <15% over next 2 quarters.

7.0 RECOMMENDATIONS & ACTION ITEMS

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7.1 Immediate Actions (Q1 2026)

Action Item	Owner	Due Date	Priority	Expected Impact
Recalibrate high-risk predictions (PD > 10%)	Model Risk Team	Feb 15, 2026	Critical	Reduce loss variance by 15%
Implement electronic AAN delivery	Compliance	Jan 30, 2026	High	100% on-time delivery
Add calibration monitoring dashboard	Data Engineering	Feb 28, 2026	High	Real-time drift detection
Reduce California concentration	Marketing	Mar 31, 2026	Medium	Lower geographic risk
Update model documentation	Data Science	Jan 20, 2026	Medium	Audit readiness

7.2 Strategic Initiatives (2026)

1. MODEL IMPROVEMENTS:

- Develop separate high-risk model for PD > 10% (Q2 2026)
- Add alternative data sources (rental payments, utility bills)
- Implement deep learning ensemble for marginal gains
- Target: AUC-ROC > 0.965 by Q4 2026

2. INFRASTRUCTURE SCALING:

- Migrate to Kubernetes for auto-scaling (Q2 2026)
- Implement multi-region active-active deployment
- Target: 5,000 requests/second capacity

3. BUSINESS EXPANSION:

- Launch in 5 new states (Q2-Q3 2026)
- Add small business lending product
- Partner with 3 new affiliate networks

4. MLOPs MATURITY:

- Implement MLflow for experiment tracking
- Automated retraining pipeline with CI/CD
- Model performance A/B testing framework

7.3 Success Metrics for Q1 2026

Metric	Target	Baseline (Q4 2025)
High-risk calibration error (PD > 10%)	< 0.50%	1.44%
AAN on-time delivery	100%	99.98%
California concentration	< 17%	18.2%
Model AUC-ROC	> 0.955	0.958
API uptime	99.95%	99.97%
Operating margin	> 78%	80.0%

8.0 APPENDICES

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APPENDIX A: Detailed Model Card

Attribute	Value
Model Name	Credit Risk XGBoost v2.1.0
Model Type	Gradient Boosted Trees
Framework	XGBoost 1.7.6
Training Date	February 1, 2025
Training Data Window	Jan 2022 - Dec 2024 (36 months)
Training Sample Size	156,432
Validation Sample Size	39,108 (20% holdout)
Number of Features	48 (8 raw + 40 derived)
Number of Trees	500
Max Depth	6
Learning Rate	0.05
Objective	binary:logistic
Evaluation Metric	AUC
Hardware Used	AWS c5.4xlarge (16 vCPU, 32GB RAM)
Training Time	4.2 hours
Inference Time (p95)	78ms

APPENDIX B: Data Dictionary (Abridged)

Full data dictionary available in Confluence (link). Key fields:

Field	Type	Description	Range/Values
person_age	Integer	Applicant age in years	18-100
person_income	Float	Annual income in USD	0-10,000,000
person_emp_length	Float	Years employed	0-60
loan_amnt	Integer	Requested loan amount	1,000-100,000
loan_int_rate	Float	Interest rate %	5.0-36.0
loan_percent_income	Float	Loan amount / annual income	0.01-1.00
cb_person_cred_hist_length	Integer	Years of credit history	0-50

APPENDIX C: Glossary of Terms

Term	Definition
AUC-ROC	Area under receiver operating characteristic curve
Calibration	Agreement between predicted probabilities and observed outcomes
Drift	Change in model or data distribution over time
ECL	Expected Credit Loss ($PD \times LGD \times EAD$)
PSI	Population Stability Index (measures feature distribution shift)
SHAP	SHapley Additive exPlanations (model interpretability)

APPENDIX D: Q4 2025 Data Files

All raw data, analysis scripts, and reproducible reports available at:

- Internal S3 bucket: `s3://credit-risk-analytics/q4-2025/`
- GitHub (private): <https://github.com/company/credit-risk-reports>

Files include:

- `applications_q4_2025.csv` (full application data)
- `performance_q4_2025.csv` (loan performance with 90-day labels)
- `model_predictions_q4_2025.parquet` (all API predictions)
- `monitoring_logs_q4_2025.json` (system metrics)

APPENDIX E: Team & Acknowledgements

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- Lead Analyst: Sarah Chen (s.chen@company.com)
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Approved by:

- Chief Risk Officer: Robert Martinez (Jan 18, 2026)
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--- END OF REPORT ---